



Priority of operations with positive integers

Task A: Multiplication, division, addition and subtraction

1) Work out the answers to the following. Ensure to show all your working out.

a) $12 + 6 \times 2$

$$12 + 12 = 24$$

c) $3 \times 10 - 24 \div 8$

$$30 - \frac{24}{8}$$

$$30 - 3 = 27$$

b) $100 - 20 \div 5$

$$100 - 4 = 96$$

2) Sam and Sofia do the same calculation differently. Whos is correct? Explain what they should have done differently.

$$10 + 15 \div 3 \times 4$$

$$10 + \frac{15}{3} \times 4$$

$$10 + 5 \times 4$$

$$10 + 20 = 30$$

Sam

$$10 + 15 \div 3 \times 4$$

$$10 + 15 \div 12$$

$$10 + \frac{15}{12}$$

$$10 + 1.25 = 11.25$$

Sofia

Sofia should have written $15 \div 3$ as a fraction and then multiply by 4.

3) Fill in the gaps with +, -, ÷ and × so that the calculations are correct.

a) $4 \quad 5 + 6 + 7 = 12$

b) $4 + 5 + 6 - 7 = 8$

c) $4 \times 5 + 6 \times 7 = 62$

d) $4 \times 5 - 6 - 7 = 7$

e) $4 \times 5 - 6 + 7 = 21$

4) Fill in the gaps with +, -, ÷ and × so that the calculations are correct.

a) $10 + 2 \div 2 - 3 = 8$

b) $10 \div 2 + 15 \div 3 = 10$

c) $10 \times 2 - 10 \div 2 + 2 \times 3 = 21$



Task B: Brackets, roots and exponents

1) Work out the answers to the following. Ensure to show your working out.

a) $24 + (6 - 2) \times 3^2$

$$24 + 4 \times 9$$

$$24 + 36 = 60$$

b) $100 - \sqrt{70 - 6} \div 2^2$

$$100 - \sqrt{64} \div 2^2$$

$$100 - 8 \div 4$$

$$100 - 2 = 98$$

2) Fill in the gaps with +, −, ÷, × and brackets so that the calculations are correct. You can use more than one pair of brackets.

a) $(4 + 5) \times (6 + 7) = 117$

b) $4 \times (5 + 6) \times 7 = 308$

c) $4 + 5 \times (6 + 7) = 69$

d) $(4 - 5) \times 6 + 7 = 1$

3) Use exactly four 4s, together with any of the operations ×, ÷, +, √, −, **exponents** (can be any value) and/or **brackets** to write calculations giving answers of: 1, then 2, then 3... up to 20. As an example, 8 has been done for you.

These are some examples of how to get each number. There are many more out there!

$$1 = 4 \div 4 + 4 - 4$$

$$2 = 4 - (4 + 4) \div 4$$

$$3 = (4 + 4 + 4) \div 4$$

$$4 = (4 - 4) \times 4 + 4$$

$$5 = (4 \times 4 + 4) \div 4$$

$$6 = (4 + 4) \div 4 + 4$$

$$7 = 4 + 4 - 4 \div 4$$

$$8 = 4 \div 4 \times 4 + 4$$

$$9 = 4 \div 4 + 4 + 4$$

$$10 = (44 - 4) \div 4$$

$$10 = (4 + 4 + 4) - \sqrt{4}$$

$$11 = (4 \div 0.4) + (4 \div 4)$$

$$12 = 4 \times (4 - 4 \div 4)$$

$$13 = (44 \div 4) + \sqrt{4}$$

$$14 = 4 \times 4 - 4 \div \sqrt{4}$$

$$15 = 4 \times 4 - 4 \div 4$$

$$16 = 4 \times 4 + 4 - 4$$

$$17 = 4 \times 4 + 4 \div 4$$

$$18 = 4 \times 4 + 4 - \sqrt{4}$$

$$19 = 4^2 + 4 - 4 \div 4$$

$$20 = 4 \times (4 \div 4 + 4)$$

